

PASSIVEGUARD AI — SIH FINAL BLUEPRINT

Frontend Technologies, Backend Architecture, Built Features & SIH Final Roadmap

Project Name: PassiveGuard AI (Unidirectional Cyber Threat Detection Platform)
SIH Domain: Cybersecurity / Air-Gapped Enclave Monitoring & Data Diode Protection
Live Production Frontend: <https://passiveguard-frontend.onrender.com>
Live Production Backend: <https://passiveguard-backend-s7vk.onrender.com>
Pytest Verification: 263 Unit & Integration Tests Passed (1 Skipped) | 100% Build Clean

1. Frontend Technologies & Component Architecture (What We Used in Frontend)

The frontend is a modern, high-performance ****Single Page Application (SPA)**** built with React 18 and Vite 6, specifically styled for high-density Security Operations Center (SOC) environments.

Category	Technology / Library	Version / Tool	Purpose & Implementation Details
Framework & Core	React JS (SPA)	18.3.1	Component-based declarative UI with custom React hooks and state management.
Build Tooling	Vite JS	6.4.3	Ultra-fast production ESM bundler, dev server, and build compiler.
React Compiler Plugin	@vitejs/plugin-react	5.1.0	Babel-based fast refresh and JSX transformation plugin for Vite.
Styling Engine	Tailwind CSS	3.4.1	Utility-first CSS framework with PostCSS and custom SOC dark theme palette.
Icon System	Lucide React	0.400.0	Scalable vector icons (Shield, Server, Radio, Activity, AlertTriangle, Cpu).
Data Visualization	Recharts	2.12.7	Interactive SVG charting (AreaChart timeline, BarChart threat distribution).
Routing Engine	React Router DOM	6.24.1	Declarative client-side SPA routing (`BrowserRouter`, `Routes`, `Route`).
HTTP Client	Axios	1.7.2	Centralized REST client with timeout configuration and HTTP error formatting.
Real-Time Stream	Native WebSocket API	Browser Native	Protocol-aware `ws://`/`wss://` client with auto-reconnect logic.

Frontend Pages Implemented:

- **SOC Dashboard (/):** Real-time active flow counts, total alerts, bandwidth, Recharts traffic rate timeline, and threat distribution charts.
- **Controlled Demo Simulator (/demo):** Interactive scenario catalog (DDoS, Recon, C2, DGA, Tunnel, TLS, Exfil, Mixed, Full Demo), execution controls, and WebSocket event log.
- **Threat Alerts Center (/alerts):** Searchable, filterable table of all recorded security findings with severity badges.
- **Threat Evidence Inspector (/alert-details):** Deep 5-tuple inspection, model confidence breakdown, explainable evidence reasons, and raw JSON payload inspector.
- **Traffic Telemetry & Analytics (/traffic):** Transport layer protocol distribution bars and historical byte throughput statistics.
- **AI Model Registry (/models):** Status registry displaying trained `joblib` model versions, dataset provenance (UNSW-NB15), and benchmark accuracy metrics.

2. Backend Technologies & System Engine (What We Used in Backend)

The backend is a high-speed Python 3.12+ asynchronous microservice built on FastAPI, Uvicorn, and scikit-learn, operating under strict zero-trust passive security constraints.

Category	Technology / Library	Version / Tool	Purpose & Implementation Details
Language & Runtime	Python	3.12 / 3.14	Core backend execution environment and numerical feature processing.
Web Framework	FastAPI	0.115.0	Asynchronous ASGI framework powering REST API endpoints and WebSocket server.
ASGI Server	Uvicorn	0.30.0	Lightning-fast ASGI web server with asyncio event loop management.
Data Validation	Pydantic & Pydantic-Settings	2.10.0	Type hint validation, Settings management, and strict schema serialization.
Machine Learning	scikit-learn	1.5.0	RandomForestClassifier model loading and real-time probability inference.
Model Serialization	Joblib	1.4.2	Efficient loading of binary <code>`joblib`</code> ML model weights.
Numerical Data	NumPy & Pandas	1.26.4 / 2.2.0	Numerical matrix manipulation and UNSW-NB15 feature matrix transformation.
Database Store	SQLite 3	Python Native	Local WAL-mode SQLite persistence for alerts and traffic snapshots (<code>`passiveguard.db`</code>).
Testing Framework	Pytest & HTTPX	8.2.0	Automated test runner executing 263 unit, integration, and AST static safety tests.

3. Built & Deployed Capabilities (What Things Are Built Now)

Component / Feature	Built Architecture	Current Operational Status	Verification / File
Passive Flow Ingestor	5-tuple FlowRecord builder tracking packets, bytes, timestamps, & transport metadata.	BUILT & WORKING	app/pipeline/engine.py
7 Threat Detectors	DDoS (ML+Stat), Recon (ML+Stat), C2 (Stat), DGA (Entropy), Tunnel (Entropy), TLS (Metadata), Exfil (Ratio).	BUILT & WORKING	app/detection/
Real ML Pipeline (UNSW-NB15)	RandomForest models trained on UNSW-NB15 DoS/Recon held-out test set (DDoS: 93.53%, Recon: 97.43%).	BUILT & WORKING	data/models/*.joblib
Multi-Signal Risk Fusion	Weighted multi-detector aggregation assigning fused risk score [0.0, 1.0] and severity tags.	BUILT & WORKING	app/risk/fusion.py
SQLite Alert & Traffic Store	WAL-mode SQLite store (`alerts` & `traffic_stats` tables) with 60s deduplication cooldown.	BUILT & WORKING	app/alerts/store.py
WebSocket Stream Server	FastAPI `/ws` streaming `alert_created`, `traffic_update`, and `state_reset` events.	BUILT & WORKING	app/api/websocket.py
Controlled Threat Simulator	Thread-safe service executing 9 scenarios (DDoS, Recon, C2, DGA, Tunnel, TLS, Exfil, Mixed, All).	BUILT & WORKING	app/demo/runner.py
SOC Dashboard SPA	React 18 frontend with Recharts timeline, alert details inspector, and model performance page.	BUILT & WORKING	frontend/src/
AST Passive Safety Checks	Static AST analyzer verifying zero socket.send(), zero active probes, zero packet emission.	BUILT & WORKING	tests/test_demo_api.py
Production Render Deployment	Deployed live on Render with CORS middleware, HTTPS/WSS support, and health checks.	BUILT & DEPLOYED	https://passiveguard-backend-s7vk.onrender.com

4. SIH Final Round Development Roadmap (What Things Are Going to Be Built)

To elevate **PassiveGuard AI** from a functional demonstration prototype to an enterprise-grade defense solution during the SIH Final, the following enhancement roadmap will be implemented:

Priority	Target Feature / Component	SIH Final Planned Architecture	Impact & Value Add
P0 (Critical)	Physical Hardware TAP & Diode Link	Integration with physical optical TAP devices and raw socket `libpcap`/`AF_PACKET` ring buffers.	Enables direct real-time capture from live 1G/10G physical data diode lines.
P0 (Critical)	NetFlow / IPFIX / sFlow Collector	Native binary UDP listener ingesting NetFlow v5/v9, IPFIX, and sFlow datagrams directly from core switches.	Allows enterprise-wide flow ingestion without requiring local packet capture agents.
P1 (High)	PostgreSQL + TimescaleDB Storage	Migrating from local SQLite to PostgreSQL with TimescaleDB time-series hyper-tables.	Supports multi-gigabyte traffic metric retention and high-throughput concurrent writes.
P1 (High)	Apache Kafka / NATS Message Bus	Asynchronous event-driven flow ingestion pipeline using NATS JetStream or Kafka topics.	Scales ingestion capacity to 100,000+ flows per second without backend bottleneck.
P1 (High)	Continuous Retraining & Drift Monitor	Automated feature distribution drift monitoring calculating Population Stability Index (PSI) to trigger retraining.	Prevents model accuracy degradation due to production network concept drift.

P1 (High)	Role-Based Access Control (RBAC)	JWT authentication with Analyst, SOC Lead, and Auditor roles regulating configuration changes.	Provides enterprise access control and compliant SOC operator audit trails.
P2 (Medium)	UI PDF Executive SOC Report Export	One-click PDF report generation button on SOC Dashboard exporting executive threat summaries.	Allows SOC managers to export instant PDF threat reports for SIH judges and auditors.
P2 (Medium)	Outbound Webhooks (Slack/Splunk)	Passive HTTP webhook dispatches pushing high-severity alert JSON payloads to external SIEMs.	Integrates seamlessly with existing SOC alert management tools (Splunk, Elastic, Slack).

5. Summary of System Readiness for SIH Final Evaluation

- **Backend Integrity:** 263 Pytest unit/integration tests passing cleanly (`263 passed, 1 skipped`).
- **Frontend Integrity:** Vite 6 SPA production bundle compiled with 0 errors (`dist/index.html`).
- **Passive Safety Guaranteed:** 100% static AST verified; 0 socket creation calls, 0 packet emissions.
- **Production Access:** Backend: <https://passiveguard-backend-s7vk.onrender.com> | Frontend: <https://passiveguard-frontend.onrender.com>

End of SIH Final Architecture & Roadmap Report — PassiveGuard AI